Agriculture: Apple Pest Management Guidelines

Powdery Mildew

Podosphaera leucotricha

Symptoms and Signs

Powdery mildew is distinguished by superficial, white powdery growth on leaves and shoots that results in the stunting and distortion of young growth. Infected fruit are stunted and russetted(/PMG/P/D-AP-PLEU-FR.001.html), and fruit set may be reduced.

Comments on the Disease

This is a major foliage disease of apples. The fungus overwinters in terminal buds(/PMG/P/D-AP-PLEU-BT.004.html) that are white, flattened, and pointed. Disease development is favored by warm days and cool, moist nights.

Management

Powdery mildew is managed primarily by pruning infected shoots during dormancy or in early spring and by applying sprays as necessary in spring to prevent buildup of the fungus and damage to the crop. Remove infected shoot tips at pruning. Chemical control of powdery mildew is done in conjunction with controls for scab. Timing and choice of material may vary from orchard to orchard.

Organically Acceptable Methods

Pruning and treatments with lime and sulfur, sulfur alone, or certain horticultural oils can be used to treat organically certified produce.

Treatment Decisions

Preferred timing is an application at pink bud(/PMG/R/S-AP-REPG-B6.002.html). If powdery mildew continues to be a problem in the orchard, apply additional treatments until terminal growth ceases.

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)

Pesticide precautions

Protect water

Calculate VOCs

Protect bees

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)

Not all registered pesticides are listed. The following are ranked with the pesticides having the greatest IPM value listed first—the most effective(/agriculture/apple/Fungicide-Efficacy-for-Apple-Diseases/) **and least likely to cause resistance are at the top of the table. When choosing a pesticide, consider information relating to the pesticide's** properties(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/) **and** application timing(/agriculture/apple/Treatment-Timings-for-Key-Diseases/), honey bees(/bee-precaution-pesticide-ratings/), **and** environmental impact(/mitigation/). **Always read the label of the product being used.**

A.	TEBUCONAZOLE/TRIFLOXYSTROBIN			
	(Adament 50WG)	4–5 oz	12	75
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Demethylation (3) and Quinone outside inhibitor (11)			
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.			
B.	TRIFLOXYSTROBIN			
	(Flint)	2–2.5 oz	12	14
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Quinone outside inhibitor (11)			
	COMMENTS: Do not apply more than 2 consecutive applications before alternating. Do not apply more than 11 oz/acre per season.			
C.	FENARIMOL			
	(Vintage SC)	9–12 oz	24	30
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Demethylation inhibitor (3)			
	COMMENTS: Do not apply more than 84 fl oz/acre per season.			
D.	MYCLOBUTANIL			
	(Rally 40WSP)	Label rates	24	14
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Demethylation inhibitor (3)			
	COMMENTS: Apply 400 gal/acre. Continue applications through the second cover spray at 7- to 10-day intervals. Use high label rate if disease was present in previous years. For application by ground equipment only.			
E.	TRIFLUMIZOLE			
	(Procure 480SC)	8–16 fl oz	12	14
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Demethylation inhibitor (3)			
F.	DIFENOCONAZOLE/CYPRODINIL			

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
	(Inspire Super)	12 fl oz	12	14
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Demethylation (3) and Anilinopyrimidine (9)			
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.			
G.	**FLUOPYRAM/TRIFLOXYSTROBIN**			
	(Luna Sensation)	5.0–5.8 fl oz	12	14
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Succinate dehydrogenase inhibitor (7) and Quinone outside inhibitor (11)			
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.			
H.	**PYRACLOSTROBIN/BOSCALID**			
	(Pristine)	14.5–18.5 oz	12	0
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Quinone outside inhibitor (11) and Carboxamide (7)			
	COMMENTS: A strobilurin/carboxyanilide fungicide.			
I.	**POTASSIUM BICARBONATE#**			
	(various)	Label rates	See label	See label
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): unknown			
J.	**MICRONIZED SULFUR#**	10–20 lb	24	0
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Multi-site contact (M2)			
	COMMENTS: May be used after bloom. Some russetting may occur in sensitive varieties if temperatures exceed 80°F.			
K.	**LIQUID LIME SULFUR#**	2 gal/100 gal dilute spray	48	0
	. . . PLUS . . .			
	WETTABLE SULFUR#	4–6 lb/100 gal	24	0
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Multi-site contact (M2)			
	COMMENTS: Do not use after bloom begins. Lime sulfur is incompatible with most other pesticides. Check before use.			
L.	**THIOPHANATE-METHYL**			

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
	(Topsin-M 70WP)	1–1.5 lb	48	1
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Methyl benzimidazole (1)			
	COMMENTS: Resistance to thiophanate methyl may develop if this material is used repeatedly. It is important to alternate this material with materials of a different chemistry. Do not apply more than 4 lb product per season.			
M.	**LIQUID LIME SULFUR#**	2–3 gal/100 gal dilute	48	0
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Multi-site contact (M2)			
	COMMENTS: Lime sulfur is incompatible with most other pesticides. Check before use. An in-season application eradicates powdery mildew.			
N.	**HORTICULTURAL OIL#**			
	(Organic JMS Stylet Oil)	1–2 gal	4	0
	MODE OF ACTION: Unknown			
	COMMENTS: Use higher rate and/or shorter spray interval when disease conditions are severe. Check with your certifier to determine other appropriate oils to use on an organically certified crop.			

‡	Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without protective clothing. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases the REI exceeds the PHI. The longer of two intervals is the minimum time that must elapse before harvest.
#	Acceptable for use on organically grown produce. Check with your certifier to make sure the selected product is permissible.
1	Group numbers are assigned by the [Fungicide Resistance Action Committee (FRAC)(http://www.frac.info/)](http://www.frac.info/) according to different modes of actions. Fungicides with a different group number are suitable to alternate in a resistance management program. In California, make no more than one application of fungicides with mode-of-action Group numbers 1,4,9,11, or 17 before rotating to a fungicide with a different mode-of-action Group number; for fungicides with other Group numbers, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action Group number.

UC PEER REVIEWED

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